
ASK DON'T TELL: REDUCING SYCOPHANCY IN LARGE LANGUAGE MODELS

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ABSTRACT

Sycophancy, the tendency of large language models to favour user-affirming responses over critical engagement, has been identified as an alignment failure, particularly in high-stakes advisory and social contexts. While prior work has documented conversational features correlated with sycophancy, we lack a systematic understanding of what provokes or prevents AI sycophancy. Here, we present a set of controlled experimental studies where we first isolate how input framing influences sycophancy, and second, leverage these findings to develop mitigation strategies. In a nested factorial design, we compare questions to various non-questions where we vary three orthogonal factors: epistemic certainty (statement, belief, conviction), perspective (I- vs user-perspective), and affirmation vs negation. We show that (1) sycophancy is substantially higher in response to non-questions compared to questions. Additionally, we find that (2) sycophancy increases monotonically with epistemic certainty conveyed by the user, and (3) is amplified by I-perspective framing. Building on this, we show that asking a model to convert non-questions into questions before answering significantly reduces sycophancy. Importantly, this effect is stronger than a simple baseline prompt asking models “not to be sycophantic”. Our work offers a practical and effective input-level mitigation that both developers and users can easily adopt.

1 Introduction

Large language models (LLMs) are increasingly used to provide advice in subjective and high-stakes domains such as health, relationships, and career decisions. In these settings, models often exhibit sycophancy by favouring user-affirming responses (e.g., using agreement, validation or flattery) rather than engaging in balanced or corrective reasoning [Sharma et al., 2025, Cheng et al., 2025]. Prior research distinguishes sycophancy from other prosocial behaviours like friendliness, showing that these dimensions interact in complex ways to shape user trust [Sun and Wang, 2025]. Recent work has characterised sycophancy as an undesirable model behaviour, particularly when it reinforces incorrect beliefs or encourages unsafe actions, and has distinguished between progressive sycophancy (agreement with correct statements) and regressive sycophancy (agreement with incorrect ones) [Fanous et al., 2025]. Existing research has primarily focused on mitigating sycophancy via explicit system-level instructions (e.g., “do not be sycophantic”). However, while potentially effective, such “black box” interventions provide only limited understanding of the factors driving model sycophancy and are not suited to derive targeted interventions. Understanding how user input framing - e.g., who speaks, what they do (ask/assert), and how (certain/hedged) - contributes to sycophantic responses could address this gap, yet very little work has explored these promising mechanisms. It remains unclear, for instance, whether sycophancy is driven by surface-level phrasing (e.g., statements vs. questions), perspective (I-perspective vs user-perspective [Wang et al., 2025, Hong et al., 2025], or inferred epistemic certainty expressed in the user input. Models’ difficulty in distinguishing beliefs from factual knowledge [Suzgun et al., 2025] makes understanding how they respond to different epistemic framing particularly important. In more naturalistic, but observational human-AI interactions settings have found correlations between conversational features and sycophancy (e.g., question frequency

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[Luettgau et al., 2025a], or perspective framing [Hong et al., 2025]). Multi-turn interactions reveal that models are particularly susceptible to sycophancy when users provide rebuttals or counterarguments, even readily agreeing with incorrect reasoning [Kim and Khashabi, 2025]. However, we currently lack a systematic understanding of how multiple input framing dimensions interact. Observational data cannot disentangle content from framing since questions and statements differ in both structure and semantic content, and controlled studies have typically examined individual framing factors in isolation. In this work, we provide a systematic, multi-dimensional examination of how user input framing causally influences sycophancy. Using content-matched prompts, we compare questions against non-questions that express the same underlying claim (e.g., “Is al dente pasta preferable to soft pasta?” vs “Al dente pasta is preferable to soft pasta”) and further vary the strength of expressed certainty (statement vs. belief vs. conviction), perspective (I-perspective vs. user-perspective) and affirmation vs. negation (cf. Figure 1A). This design allows us to test whether sycophancy is primarily driven by inferred user certainty rather than by topical content. Using two rubric-based LLM-as-a-judge grader models to evaluate sycophancy, we analysed behaviour of three frontier models using hierarchical Bayesian statistical modeling. We find that questions elicit substantially less sycophantic responses than matched non-questions, and that within non-questions, sycophancy increases monotonically with expressed certainty (convictions > beliefs > statements) and is amplified by I-perspective framing (Section 3.1). These effects persist after controlling for response length and model-specific differences. Building on these findings, we show that rephrasing non-questions as questions yields a large reduction in sycophancy, outperforming explicit no-sycophancy instructions (Section 3.2). For completeness, we also evaluate reframing I-perspective claims into user-perspective claims, which produces a smaller effect (Section 3.3). We explore sycophantic responses across different models and in response to different topics in the user inputs, revealing significant heterogeneity and modulation of sycophancy (Section 3.4).

This paper makes two contributions:

1. We provide controlled experimental evidence that user input framing, particularly question vs. non-question form and expressed certainty in non-question inputs, drives LLM sycophancy.
2. We demonstrate simple, easily deployable prompt-based mitigations (e.g., “rephrase the input as a question before answering”) that significantly reduce sycophantic behaviour across models, to a greater extent than black box prompts (e.g., “don’t be sycophantic”).

Our work suggests that input-level changes can be used as an effective sycophancy mitigation strategy for both model developers (e.g., by adding a rewriting instruction to the system prompt), and users (e.g., by carefully thinking about how they frame what they write when interacting with chatbots).

2 Related work

2.1 Sycophancy in language models

Recent work has identified sycophancy as a form of misalignment [Sharma et al., 2025, Cheng et al., 2025]. Empirical studies have shown that models may adapt their responses to inferred user preferences even when doing so conflicts with factual accuracy or critical reasoning, particularly in subjective or normative domains [Chen et al., 2024]. Recent work has begun to distinguish sycophancy from related but distinct prosocial behaviours such as friendliness, revealing that these dimensions interact in complex ways and differentially affect user perceptions of authenticity and trust [Sun and Wang, 2025]. This behaviour has been linked to both reinforcement learning from human feedback (RLHF) and instruction-following objectives that implicitly reward user satisfaction. Shapira et al. [2026] formally demonstrate that RLHF causally amplifies sycophancy through a covariance mechanism between endorsing user beliefs and learned rewards. Prior work has further distinguished between *progressive sycophancy*, in which agreement coincides with correct or benign outcomes, and *regressive sycophancy*, in which agreement reinforces incorrect beliefs or harmful actions [Fanous et al., 2025]. This distinction is particularly important given that language models cannot reliably distinguish belief from knowledge and fact [Suzgun et al., 2025], meaning they may agree with user statements without appropriately weighing their epistemic status. While this distinction clarifies why sycophancy can be difficult to detect and mitigate, existing studies largely focus on identifying its presence rather than isolating its causes and suggesting mitigations. Moreover, multi-turn interaction studies reveal that sycophancy can be dynamically triggered or amplified: models show increased susceptibility when users provide rebuttals or counterarguments, readily agreeing even with incorrect reasoning, particularly when feedback is casually phrased or includes detailed (but flawed) justification [Kim and Khashabi, 2025].

2.2 Input framing

Several studies have explored how specific linguistic cues influence model behaviour. User inputs framed in I-perspective (e.g., “I believe...”) have been shown to increase agreement rates in comparison to user-perspective framing (e.g., “they believe...”) [Wang et al., 2025], while prompting models to adopt an external perspective can reduce certain forms of bias [Hong et al., 2025]. Related work on stance-taking and social mirroring suggests that models may infer epistemic commitment, confidence, or intent from surface-level linguistic markers, adjusting their responses accordingly [Pickering and Garrod, 2004, Perez et al., 2022]. However, these findings are typically derived from correlational analyses or single-condition prompt interventions. As a result, it remains unclear whether observed effects are driven by semantic differences in content, pragmatic differences in conversational intent, or inferred user certainty. Our work extends prior work by systematically teasing apart the influence of these different factors, using content-matched prompts that isolate framing effects while holding propositional content constant.

2.3 Prompt-based mitigation strategies

A growing body of work investigates prompt-level interventions to steer model behaviour, including instructions to avoid agreement, adopt critical reasoning, or ask clarifying questions. While such approaches can reduce overt sycophancy, they often rely on explicit system-level directives (“do not agree with the user”) and operate as a black box, limiting our understanding of causal drivers and the ability to design targeted interventions. In contrast, recent work has suggested that indirect prompt transformations, such as self-asking, rephrasing, or intermediate representations, can alter model behaviour without explicitly constraining the output [Hong et al., 2025]. Our mitigation strategies build on this idea by manipulating user input framing rather than output constraints. Importantly, we evaluate these interventions using matched control conditions, allowing us to distinguish genuine framing effects from generic prompt-induced shifts.

3 Results

We constructed a controlled prompt set of 440 variants across four domains (hobbies, social relationships, mental health, and medical topics; illustrated in Figure 4A), systematically varying input framing while holding propositional content constant. For each base question (e.g., “Do couples with the same mother tongue have stronger bonds?”), we generated content-matched non-questions varying in epistemic certainty (statement, belief, conviction), perspective (I-perspective “I” vs. user-perspective “the user”), and affirmation vs. negation (cf. Figure 1A). We evaluated three frontier language models (GPT-4o, GPT-5, and Sonnet-4.5) across all conditions, generating responses constrained to 150–200 words.

Sycophancy was assessed using a rubric-based LLM-as-a-judge approach with two grader models (GPT-5 and Sonnet-4.5), scoring five facets of sycophantic behaviour (excessive agreement, flattery, avoiding disagreement, user preference alignment, and validation seeking) on a 0–3 scale per facet (total range: 0–15). For the full rubric cf. Appendix B.1. For mitigation studies, we evaluated question-reframing mitigations (converting non-questions to questions; illustrated in Figure 2A) and perspective-reframing mitigations (converting I-perspective to user-perspective; illustrated in Figure 3A), comparing these to direct responses and a no-sycophancy baseline instruction. All analyses used Bayesian Generalised Linear Models with ordered-logistic likelihoods, controlling for topic domain, model, grader, and response length. Full methodological details are provided in the Methods section 7.